

Azure Synapse Analytics – Interview Q&A; Pack

Q1. What is Azure Synapse Analytics?

A unified cloud analytics platform combining data warehousing, big data, data integration, and orchestration.

Q2. Difference between Dedicated SQL & Serverless SQL?

Dedicated SQL: Provisioned, MPP, high performance.

Serverless SQL: Pay-per-query, ad-hoc analytics.

Q3. What is MPP architecture?

Massively Parallel Processing: Distributes data + queries across compute nodes to process workloads simultaneously.

Q4. Explain Synapse distributions.

Hash: Performance for joins.

Round-Robin: Quick load, no distribution key.

Replicate: Small dimension tables.

Q5. What is a Columnstore Index?

A compressed, column-wise storage optimized for analytics workloads.

Q6. What is PolyBase?

Parallel data ingestion system used to load data efficiently into Dedicated SQL Pool.

Q7. Difference between Spark and SQL Pools?

Spark: Big data, ML, unstructured processing.

SQL Pools: Structured data, BI, warehousing.

Q8. What is Delta Lake?

An ACID-compliant storage layer over Parquet supporting updates, deletes, merges, and versioning.

Q9. What is the Lakehouse architecture?

A unified architecture combining Data Lake + Warehouse using Delta Lake as the storage backbone.

Q10. How does Synapse ensure security?

RBAC, managed identity, private endpoints, TDE, SSE, firewall, least privilege access.

Q11. How is data loaded into Synapse?

Using PolyBase, COPY INTO, Dataflows, Pipelines, Spark.

Q12. What are Materialized Views?

Precomputed query results improving performance for complex aggregations.

Q13. What is CTAS?

Create Table As Select – fastest way to transform data.

Q14. What is the purpose of Data Explorer Pool?

Kusto engine for log & telemetry analytics.

Q15. What is OPENROWSET?

A serverless SQL function to query files directly from ADLS.